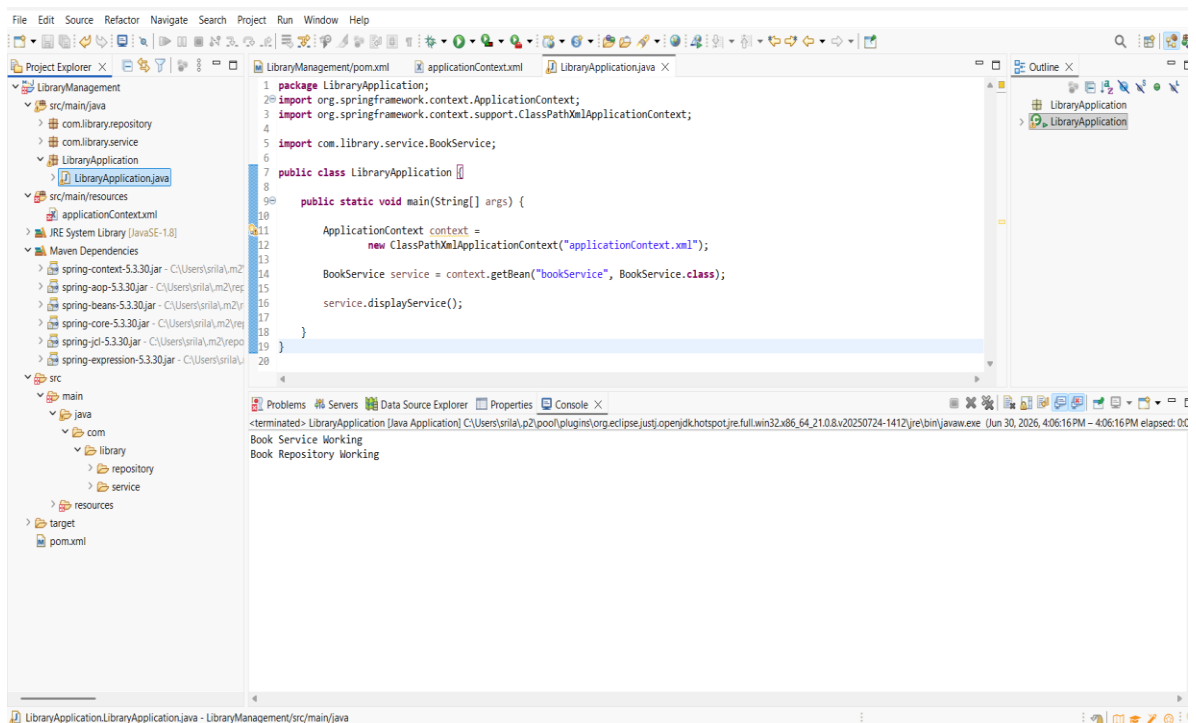


Spring Core and Maven

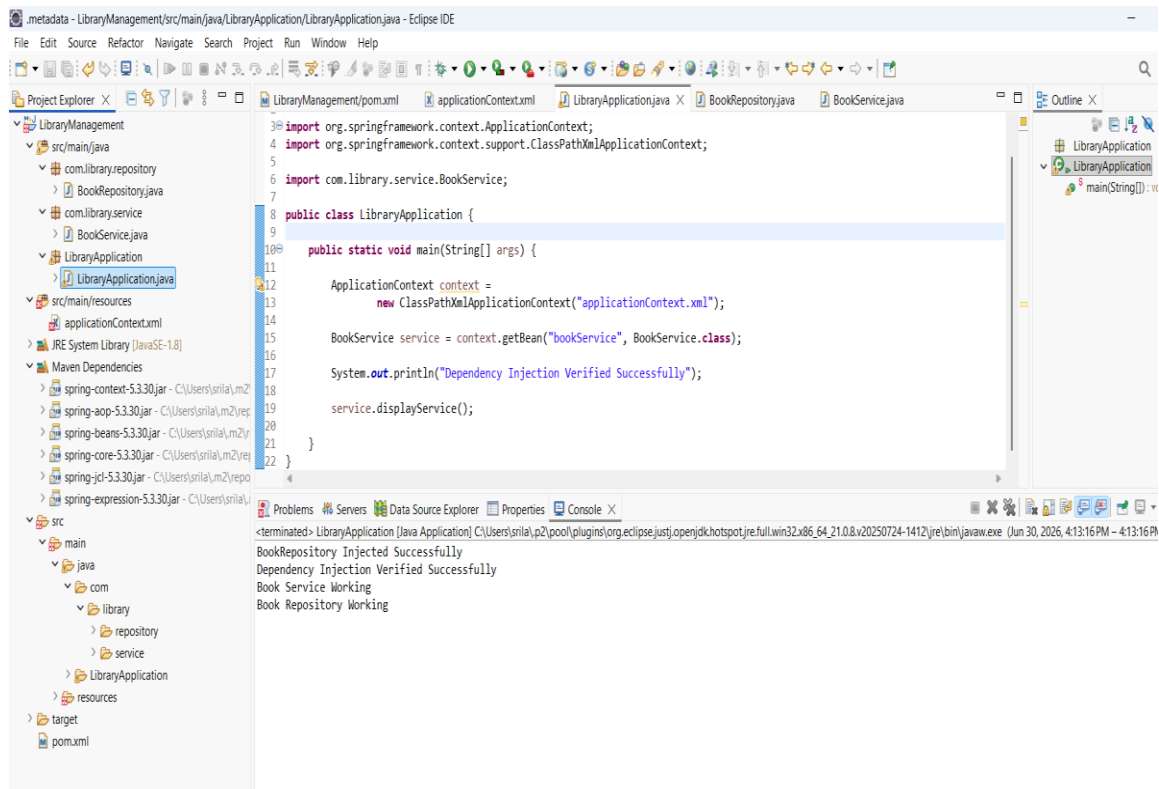
Exercise 1: Configuring a Basic Spring Application

In this exercise, a Maven-based Spring project named LibraryManagement was created. The Spring IoC container was configured using applicationContext.xml, and beans for BookService and BookRepository were defined and loaded successfully.



Exercise 2: Implementing Dependency Injection

Dependency Injection was implemented using Setter Injection. BookRepository was injected into BookService through the XML configuration using the <property> tag. The application verified successful dependency injection through the console output.



```
.metadata - LibraryManagement/src/main/java/LibraryApplication/LibraryApplication.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer X
LibraryManagement
  src/main/java
    com.library.repository
      BookRepository.java
    com.library.service
      BookService.java
      LibraryApplication.java
  src/main/resources
    applicationContext.xml
  JRE System Library [JavaSE-1.8]
  Maven Dependencies
    spring-context-5.3.30.jar - C:\Users\sri1a\m2
    spring-aop-5.3.30.jar - C:\Users\sri1a\m2\rep
    spring-beans-5.3.30.jar - C:\Users\sri1a\m2\rep
    spring-core-5.3.30.jar - C:\Users\sri1a\m2\rep
    spring-jcl-5.3.30.jar - C:\Users\sri1a\m2\rep
    spring-expression-5.3.30.jar - C:\Users\sri1a\m2\rep
  src
    main
      java
        com
          library
            repository
            service
          LibraryApplication
      resources
      target
      pom.xml

LibraryManagement/pom.xml
applicationContext.xml
LibraryApplication.java
BookRepository.java
BookService.java

Outline X
LibraryApplication
  LibraryApplication
    main(String[]) : void

38 import org.springframework.context.ApplicationContext;
4  import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6  import com.library.service.BookService;
7
8  public class LibraryApplication {
9
10
11
12  public static void main(String[] args) {
13
14      ApplicationContext context =
15          new ClassPathXmlApplicationContext("applicationContext.xml");
16
17      BookService service = context.getBean("bookService", BookService.class);
18
19      System.out.println("Dependency Injection Verified Successfully");
20
21      service.displayService();
22  }
23 }
```

```
<terminated> LibraryApplication [Java Application] C:\Users\sri1a\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.8\v20250724-1412\jre\bin\javaw.exe (Jun 30, 2026, 4:13:16 PM)
BookRepository Injected Successfully
Dependency Injection Verified Successfully
Book Service Working
Book Repository Working
```

Exercise 4: Creating and Configuring a Maven Project

The Maven project was configured by adding Spring Context, Spring AOP, and Spring WebMVC dependencies in pom.xml. The Maven Compiler Plugin was configured for Java 1.8, and all required libraries were downloaded successfully.

